

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-V	20.00	21.05
2	Module-I	20.00	19.74
3	Module-II	20.00	19.74
4	Module-III	20.00	19.74
5	Module-IV	20.00	19.74
Weightage representation okay : Weightage representation exceeded more than 5% :Weightage representation exceeded more than 10%			

CSIT617

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Python for Data Science

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.	(a) How do you create and store strings in Python? List any 5 string functions.	(3)	Delete
	(b) Explain the use of numpy for creating and managing arrays.	(3)	Delete
Q2.	(a) What is a Lambda function? Provide a syntax example.	(3)	Delete
	(b) Describe the try-except-else-finally block structure.	(3)	Delete
Q3.	Compare and contrast Data Science and Data Analytics.	(6)	Delete
Q4.	(a) Define Random Variables in the context of inferential statistics.	(3)	Delete
	(b) What is the difference between Normal and Exponential continuous distributions?	(3)	Delete
Q5.	Explain the role of Subplots and their properties in matplotlib for data visualization.	(6)	Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.	(a) Discuss the various loop control statements (for, while) and the use of break and continue.	(5)	Delete
	(b) How are lists, tuples, and dictionaries passed as arguments to functions?	(5)	Delete
Q7.	(a) Explain the process of plotting Sine and Cosine curves using matplotlib.	(5)	Delete
	(b) Describe how to extract components from an HTML document using Beautiful Soup.	(5)	Delete
Q8.	(a) Discuss Data Wrangling and the importance of handling outliers.	(5)	Delete
	(b) Explain the concepts of Slicing, Dicing, and Filtering data in Python.	(5)	Delete

Section - C : Compulsory question

Q9.	(a) Describe how to read and write JSON files in Python.	(4)	Delete
	(b) Discuss the measures of Shape and Symmetry (Skewness and Kurtosis) in data.	(4)	Delete
	(c) Explain the difference between Population and Sample in the context of statistics.	(4)	Delete
	(d) Explain how to visualize data using Histograms and Pi charts in Python.	(4)	Delete